# Chelse Swoopes

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Research Interests

Human-Computer Interaction; Human-AI Interaction and Collaboration; Interface Support for AI-Resiliency; Theories of Human Concept Learning; Theories of Human Cognition; Theory-Driven Interface Design

## Education

**Ph.D.**, Computer Science — Harvard University, Cambridge, MA, USA — Sept 2021 - May 2027
Advisor: Assist. Prof. Elena L. Glassman
Member of The Variation Lab

**M.Sc.**, Computer Science — Harvard University, Cambridge, MA, USA — Sept 2021 - May 2024
Advisor: Assist. Prof. Elena L. Glassman
Member of the Variation Lab

**M.Sc.**, Electrical Engineering, Carnegie Mellon University, Pittsburgh, PA, USA — Aug 2016 - Dec 2017
Advisor: Prof. Lorrie F. Cranor
Member of the CyLab Usable Privacy and Security Lab (CUPS)

**B.Sc.**, Mathematics, Mississippi State University, Starkville, MS, USA — Aug 2013 - May 2016
*Summa Cum Laude*

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## Publications

### *Peer-Reviewed Conference Publications*

CorpusStudio: Surfacing Emergent Patterns in a Corpus of Prior Work while Writing — 2025
Hai Dang*, **Chelse Swoopes***, Daniel Buschek, Elena L. Glassman
*In Proceedings of the CHI Conference on Human Factors in Computing Systems, CHI'25, 2025.*

Chainforge: A Visual Toolkit for Prompt Engineering and LLM Hypothesis Testing — 2024 🏆
Ian Arawjo, **Chelse Swoopes***, Priyan Vaithilingam*, Martin Wattenberg, Elena L. Glassman.
*In Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems,*

*2024.*
*Best Paper Honorable Mention Award*

Supporting Sensemaking of Large Language Model Outputs at Scale 2024 🏆
Katy Ilonka Gero, **Chelse Swoopes**, Ziwei Gu, Jonathan K. Kummerfeld, Elena L. Glassman.
*In Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems, 2024.*
*Best Paper Honorable Mention Award*

An empirical analysis of data deletion and {Opt-Out} choices on 150 websites 2019
Hana Habib, Yixin Zou, Aditi Jannu, Neha Sridhar, **Chelse Swoopes**, Alessandro Acquisti, Lorrie Faith Cranor, Norman Sadeh, Florian Schaub.
*In Proceedings of the Fifteenth Symposium on Usable Privacy and Security (SOUPS 2019), 2019.*

Turtles, Locks, and Bathrooms: Understanding Mental Models of Privacy Through Illustrations 2018 🏆
Maggie Oates, Yama Ahmadullah, Abigail Marsh, **Chelse Swoopes**, Shikun Zhang, Rebecca Balebako, Lorrie Faith Cranor
*In Proceedings of the Privacy Enhancing Technologies Symposium. 2018.*
*Andreas Pfitzmann Best Student Paper Award*

User Behaviors and Attitudes Under Password Expiration Policies 2018
Hana Habib, Pardis Emami Naeini, Summer Devlin, Maggie Oates, **Chelse Swoopes**, Lujo Bauer, Nicolas Christin, Lorrie Faith Cranor.
*In Proceedings of the Fourteenth Symposium on Usable Privacy and Security (SOUPS'18). 2018.*

'It's not actually that horrible': Exploring adoption of two-factor authentication at a university. 2018
Jessica Colnago, Summer Devlin, Maggie Oates, **Chelse Swoopes**, Lujo Bauer, Lorrie Cranor, Nicolas Christin.
*In Proceedings of the ACM SIGCHI Conference on Human Factors in Computing Systems (CHI'18). 2018.*

## *Peer-reviewed Workshop Publications*

The Impact of Revealing Large Language Model Stochasticity on Trust, Reliability, and Anthropomorphization 2024
**Chelse Swoopes**, Tyler Holloway, Elena L. Glassman.
*Trust and Reliance in Evolving Human-AI Workflows (TREW) Workshop at CHI, 2024.*

Reverse Sketching
Tyler Holloway, **Chelse Swoopes**, Ian Arawjo, Hila Peleg, Elena L. Glassman. 2023
*PLATEAU, 2023.*

Accurate, explainable, and private models: providing recourse while minimizing training data leakage 2023
Catherine Huang*, **Chelse Swoopes***, Christina Xiao*, Jiaqi Ma, Himabindu Lakkaraju.
ICML Workshop on New Frontiers in Adversarial Machine Learning, 2023

Information Sharing, Security Models, and Cyber Defense. *[Shared by request]* 2019
Michael Jerge*, **Chelse Swoopes***, Mike Ridge, Joe Brule.
*In Proceedings of the Coordinating Attack Response at Internet Scale (CARIS) Workshop, 2019*

### *Posters*

AI-Resiliency in Legal Work: A Heuristic Evaluation of LLM-Based Workflow Support Tools in Research and Practice 2026
**Chelse Swoopes**, Elena L. Glassman, Aileen Nielsen.
*In Proceedings of the ACM Symposium on Computer Science and Law (CS&Law 2026).*

Interface Design to Support Legal Reading and Writing: Insights from Interviews with Legal Experts. 2026
**Chelse Swoopes**, Ziwei Gu, Elena L. Glassman.
*In Proceedings of the ACM Symposium on Computer Science and Law (CS&Law 2026).*

Judges are Vulnerable to AI Influence; Interface Design Can Help. 2026
Aileen Nielsen, **Chelse Swoopes**, Elena L. Glassman
*Accepted to ACM Symposium on Computer Science and Law (CS&Law 2026).*

"I Tried to Buy Drugs Online Using Crypto but Failed": Understanding the Usability & Perceptions of Bitcoin 2018
**Chelse Swoopes**, Daniel Stiffler, Saksham Chitkara, David Edelstein, Lorrie Faith Cranor.
*In Proceedings of the Fourteenth Symposium on Usable Privacy and Security (SOUPS 2018).*

It's Not Actually that Horrible: Exploring Adoption of Two-Factor Authentication at a University 2018
Jessica Colnago, Summer Devlin, Maggie Oates, **Chelse Swoopes**, Lujo Bauer,

Lorrie Faith Cranor, and Nicolas Christin.
*In Proceedings of the Fourteenth Symposium on Usable Privacy and Security (SOUPS 2018).*

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## Awards and Fellowships

### *Scientific Research*

**Best Paper Award**, PETS 2018 for "Turtles, Locks, and Bathrooms: Understanding Mental Models of Privacy Through Illustrations." 2018

**Best Paper Honorable Mention Award**, ACM CHI 2024 for "Supporting Sensemaking of Large Language Model Outputs at Scale." 2024

**Best Paper Honorable Mention Award**, ACM CHI 2024 for "Chainforge: A Visual Toolkit for Prompt Engineering and LLM Hypothesis Testing." 2024

### *Fellowship*

Siebel Scholar, Thomas and Stacey Siebel Foundation 2026

Nicole A. Chen and Karina A. Chen Graduate Student Research, Harvard University 2021-2022

### *Industry Recognition*

Outstanding Contributor Award, The Charles Stark Draper Laboratory 2019
*Awarded in <2 years for technical depth, project strategy and delivery, and professionalism.*

2nd Place in Biometric Identification, EMC Corporation Cyber Case Competition 2016
*Researched and presented on improving and innovating biometric methods for identification*

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## Service & Outreach

### *Peer Reviewing*

CHI (2018, 2019, 2024, 2025, 2026), UIST (2025), PLATEAU (2023)

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## Teaching & Mentoring Experience

**Teaching Assistant**, Fairness and Privacy: Perspectives of Law and Probability (CS 126) — Joint course with SEAS (Computer Science) and Harvard Law School, Harvard University. Fall 2022 and 2023
Instructor of Record: Prof, Cynthia Dwork, Prof. Martha Minow
Responsibilities: Developed project assignments and exams, led weekly teaching sessions and office hours, and taught students to engage in meaningful cross-disciplinary communication and collaboration between law and computer science.

**Lead Mentor**, Bagley College of Engineering: Chevron Engineering Transfer Program, Mississippi State University, Starkville, MS, USA 2015-2016
Responsibilities: Facilitate pathways for students transitioning from community college or other institutions of higher learning, fostered a sense of community among transfers and students currently matriculating within engineering, introduced students to available resources that would allow for increased student engagement, academic enrichment, and improved rates of retention

## Research Experience

**Graduate Research Assistant**, Variation Lab, Department of Computer Science, Harvard University. 2021 - Present
Advisor: Assist. Prof. Elena L. Glassman
*[Programming Languages Lab, Fall 2021-Spring 2023]*

**Research Associate**, CyLab Security and Privacy Institute, Carnegie Mellon University, Pittsburgh, PA 2018
Capacities: Researcher, mentor to M.Sc. and undergrad students
Responsibilities: Planned, designed, and conducted usable security and privacy assessments with human subjects, Analyzed data by using qualitative & quantitative research methods, Co-authored peer-reviewed publications

**Graduate Research Assistant**, CUPS Lab, Department of Computer Science-CyLab, Carnegie Mellon University. 2017
Advisor: Prof. Lorrie Cranor

**Research Associate**, Center for Advanced Vehicular Systems, Mississippi State University, Starkville, MS, USA May-July 2016
Principal Investigator: Dr. Daniel Carruth

**Undergraduate Research Assistant**, The National Forensics Training Center and

Center for Computer Security Research, Mississippi State University, Starkville, MS, USA — June-Aug. 2013
Advisor: Prof. Dave A. Dampier
Responsibilities: Researched various ways that cybercrime investigators handle recovered data, developed a tool that recovered evidence of internet history on a machine's hard-drive using Python and SQL, managed project tasks, changes, and interventions to achieve project outputs, presented research at MSU's Summer 2013 Undergraduate Research Symposium

**Undergraduate Researcher,** Broadening Participation in Computing Program, — June-July 2012
Responsibilities: • Developed a program to control the movement of a remote control vehicle using Spyder, conducted research on valuable leadership skills and ethics in computing

## Industry Experience

**Sr. Systems Security Engineer**, The Charles Stark Draper Lab, Cambridge, MA — 2018 - 2022
*Technical Director*
Responsibilities: Staffed personnel with programs based upon skill set needed to successfully drive program to completion, Improved task management, internal and customer communication, and created training material, Responded to DoD request for information and request for prototype proposals, Maintained awareness of policies and technologies relevant to program's business & critical capabilities

*Task Lead*
Responsibilities: Led a team of 5 in the development of model based engineering (MBE) technologies, Maintained significant operational tempo and briefed technical content that exceeded customer expectation, Served as corporate liaison in the development of a consortium of collaborators and potential customers

*System Qualification Individual Contributor*
Responsibilities: Derived system level requirements from customer driven requirements, Evaluated MagicDraw system models and Python test scripts for verification of meeting system requirements or test descriptions

*Program Protection Individual Contributor*
Responsibilities: Conducted security improvement studies on customers' proprietary and open architecture platforms, Ensured DoD critical systems

maintain an Authorization to Operate by completing NIST and customer based risk management framework assessments, Created and consulted on the development of information assurance and system security plans for DoD customers, Contributed to technical working group as a cyber security representative for Draper, Coordinated and led a multi-organization tabletop exercise that was delivered to the OSD

*Principal Investigator, Draper High School Summer Internship Program* (2019)
Responsibilities: Provided technical guidance during the build of a water test kit using Raspberry Pi's and sensors, Actively mentored 4 students with a $25k budget over 6 weeks

*Advocate for Diversity, Equity, and Inclusion in the Workplace*
Responsibilities: Inaugural Co-Lead of the Early Career Employee Resource Group tasked (Led 3 process action teams composed of 6 staff members with a labor and M&S budget, Created community building, professional development, and networking resources and programs, Consulted executive strategic planning committees on topics such as organization culture), Active member of the Black Inclusion Network at Draper (BIND) Employee Resource Group, Active member of the Diversity & Inclusion Co-Lead Selection Committee (tasked with defining candidate criteria and selection process, interviewing prospective candidates, and briefing executives on candidate recommendation)

*2020 Outstanding Contribution Award Recipient*
Recognizes employees who produce superior results that add value to the organization through: significant achievement of projects or organizational goals, increasing operational efficiency, enhancing business processes, or demonstrating exceptional behavior in mentoring leadership or service.

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## Selected Talks and Panels

Carnegie Mellon University, CyLab Usable PrIvacy and Security Lab Seminar — 2018
Presentation: "I Have Three That I Rotate": Behaviors and Attitudes Toward Password Expiration

### *Panels*

Harvard University, CS290: Seminar on Effective Research Practices & Academic Culture, Panel on Managing Advising Relationships — 2022

Carnegie Mellon University, Preparing for STEM-based Graduate Programs 2018

Carnegie Mellon University, Provost Diversity & Inclusion Roundtable
Provided student perspective, feedback, and strategy recommendations on university D&I mission and priorities 2018

## Selected Press & Exhibitions

*Press*

"*Siebel Scholars Foundation Announces Class of 2026*," Business Wire 2026

Aileen Nielsen, "Law is vulnerable to AI influence; interface design can help," The Berkman Klein Center for Internet & Society, 2025 2025

Leah Burrows, "SEAS at CHI SEAS teams win Best Paper, honorable mentions at international conference on Human-Computer Interaction", Harvard SEAS News, 2024 2024

Daniel Tkacik, "Opting out of data use is hard, but it doesn't have to be", CyLab News, 2019 2019

## Certifications

**No Magic, Inc.** SysML Intensive with MBSE using Cameo Systems Modeler Jan. 2020

**Department of Defense Anti-Tamper Executive Agent** Anti-Tamper Policy, Technology and Application Sept. 2019

## Skills

**Application Software**:
MagicDraw, STIG Viewer, Mathematica, MATLAB, Maple, FTK Imager, Autopsy, Opanda IExif, TAMS Analyzer, Atlas.ai,

**Programming Languages**:
Python, C, R

## Volunteering

**Judge**, Holmes Innovation School STEM Fair, Boston, MA, USA

**Distributer**, Greater Boston Food Bank, Boston, MA, USA

**Distributer**, Cradles to Crayons, Boston, MA, USA

2025

**Operation Snow Sports Ski Instructor**, Youth Enrichment Services, Boston, MA, USA

2018, 2021

Responsibilities: teach youth (grades 2-8) fundamentals of skiing, promote and demonstrate safety protocols, and encourage youth to be inspired through the challenges of learning

2018, 2021

2019-2020

**Community Service Chair**, Black Graduate Student Association, Carnegie Mellon University, Pittsburgh, PA, USA
Responsibilities: Organized >4 events to connect CMU with Greater Pittsburgh students and residents

**Camp Counselor**, The Bulldog Bytes Aspire IT Camp, Mississippi State University, Starkville, MS, USA

2016-2017

Advisor:  Prof. Sarah Lee
Responsibilities: Mentored on self-worth and the importance of women in STEM, taught middle school girls how to program using Scratch, used Scratch to program a Finch robotic device, assisted participants in developing a communication tool for educating their peers about identity theft

2013

## Affiliations

National Society of Black Engineers
Society of Women Engineers
Omicron Delta Kappa National Leadership Honor Society
Association for Computing Machinery
Mississippi State University Alumni Association
Carnegie Mellon University Alumni Association
National Society of Black Engineers - Boston Professionals Chapter
Harvard Alumni Association
Harvard Black Alumni Society